

Rachelle Bassen
CS-499 Capstone Professional Self-Assessment

The final Capstone project for the SNHU BS in Computer Science degree reflects my dedication to challenge, creativity and curiosity when building collaborative environments that enable diverse audiences to support organizational decision making in the field of computer science. The enhancements of two artifacts produced in previous courses demonstrate my commitment to staying current in technology. This was supported by the successful deployment of the WNC-NFA Chapter Website built on React with Vite frontend on an Express/Node.js backend with PostgreSQL database. The chosen stack for the WNC-NFA website aligns with emerging trends in website UI design built on component-based architecture, leading to straightforward and user-friendly navigation.

The full-stack development for the WNC-NFA Chapter Website was facilitated by previous knowledge acquired in CS-305 and CS-340. The assignments in those courses helped me understand secure coding practices and database management such as password hashing, user authentication, and running dependency checks for vulnerabilities. The preliminary exposure to database management was especially helpful in learning how to integrate a new database like PostgreSQL into my Capstone project. Awareness of modern development lifecycles used in industry software development such as Agile, which were shaped in CS-250, helped me with building and testing each project enhancement iteratively.

Computer science fields depend on critical skills to maintain accountability and secure implementation of software around the world. As a current college professor, I have an extensive history of communicating with stakeholders to develop program outcomes. It is sometimes challenging to glean stakeholder needs, but those barriers can be overcome by asking the right questions and maintaining regular communication throughout the project to ensure that the buildout aligns with their expectations. Working collaboratively with team members also ensures that the project stays on track and duties are assigned fairly and appropriately. I also have experience working on college program assessments with a group of colleagues. Acknowledging different skillsets and communicating regularly on progress throughout the project are important to ensure that project requirements are fulfilled. Honesty and transparency about project challenges promotes learning and should not be viewed as faults. These skills were supported throughout the Computer Science degree in the myriad of

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discussion posts required across all courses. I also supported my peers in the SNHU Student Discord group by providing feedback and resources for different courses.

Designing and delivering professional-quality written and visual communications that are coherent, technically sound, and appropriately adapted to specific audiences has been a consistent focus throughout the program. The WNC-NFA Chapter Website is an example of audience-adapted communication because it presents the union information and survey results in a clear, accessible format for faculty members who may not be technologically savvy. The BS degree in Computer Science at SNHU strengthened my skills in data structures and algorithms, software engineering and database, and security. Technical writing assignments, code documentation, and the development of this ePortfolio further required me to communicate complex technical concepts clearly to both technical and non-technical audiences.

The artifacts presented in this ePortfolio collectively demonstrate a full range of computer science skills that I achieved throughout the SNHU BS in Computer Science program. Each artifact addresses one of the three enhancement categories required for the CS-499 Capstone project: Software Design and Engineering, Algorithms and Data Structures, and Databases.

The foundation for Enhancements One and Three is the CS-465 Full-Stack Final Project: a travel booking website called “Travlr Getaways”. This application was built on a MEAN (MongoDB-Express-Angular-Node.js) stack, but recompiled into a React-Express-Node.js-PostgreSQL stack as a proficient demonstration of a modern full-stack website. The first enhancement, Software Design and Engineering, involved migrating the Angular frontend to React with Vite and redesigning the application as the WNC-NFA Chapter Website. This enhancement demonstrates application in modern frontend development which is built on responsive UI design using Material UI and full-stack deployment in a local environment using Express/Node.js and publicly displayed through GitHub Pages and Render. The full-stack website application demonstrates my ability to use well-founded and innovative techniques, skills and tools in computing practices for the purpose of implementing computer solutions that deliver value and accomplish industry-specific goals.

Enhancement One also served as an opportunity to showcase a security mindset by mitigating design flaws in anticipation of adversarial exploits in software architecture and designing to expose potential vulnerabilities, and ensuring privacy and enhanced security of data and resources through consistent class structures, clear variable naming conventions, thorough debugging, and sufficient code documentation throughout the development process.

The second enhancement, Algorithms and Data Structures, focused on the survey results. Survey responses are stored in the PostgreSQL database, then retrieved and sorted using a quicksort algorithm. The data was then compiled into aggregated graph data for frontend display. This demonstrates a practical application of designing and evaluating computing solutions that solve a given problem using algorithmic principles and computer science practices and standards appropriate to its solution, while managing the trade-offs involved in design choices. For example, the design trade-off of transforming the database responses into a structured graph for frontend display was effectively managed by separation of concerns across controller, model and service functions.

Migrating MongoDB to PostgreSQL was a showcase of the third enhancement, Databases. The PostgreSQL database was used to store user authentication and survey data responses. This enhancement demonstrates an ability to use well-founded and innovative techniques, skills, and tools in computing practices for the purpose of implementing computer solutions that deliver value and accomplish industry-specific goals several skills such as relational database design, writing and executing PostgreSQL queries for successful data retrieval and aggregation, and implementing secure cloud database infrastructure and API integration between the backend and database components.

The three enhancements integrated within this project provide evidence of my abilities to independently design, build, and deploy a complete full-stack web application. The technical skills and professional values that I developed throughout the SNHU BS degree in Computer Science are also demonstrated from the successful completion of this Capstone project.